

Emertex Smart Edge-AI EV Charging Optimizer

Predictive load management and execution stack
for constrained infrastructure.

Feeder Capacity (64A) < Simultaneous Demand (3x 32A)

Electrical Hazard



Simultaneous full-current charging leads to absolute feeder overload, risking a tripped breaker or electrical fire.

Thermal Hazard



Sustained high-current draw causes connector temperatures to rise rapidly, risking permanent cable damage.

Capacity Gridlock



Without dynamic control, a newly arriving EV encounters zero electrical headroom and cannot charge.

Required: A sub-second control loop to sense, predict, and react to physical constraints autonomously.



Sub-second control

Control decisions are made locally every single second without network round-trips.



Tolerant of offline conditions

No critical safety loop relies on an active internet connection.

Why Edge AI over Cloud AI?



Zero cloud inference bills

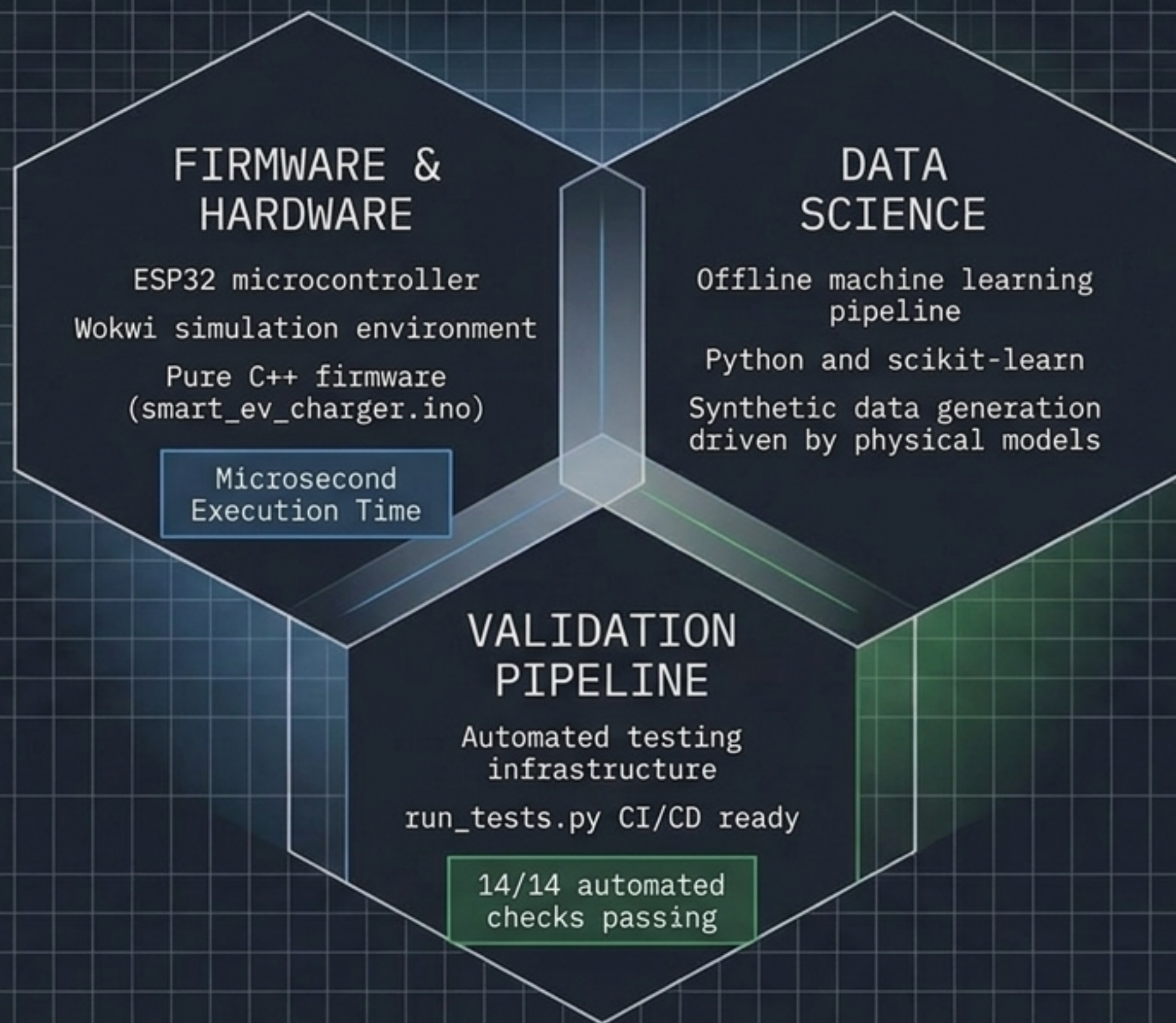
Compute power scales infinitely with physical hardware deployment.



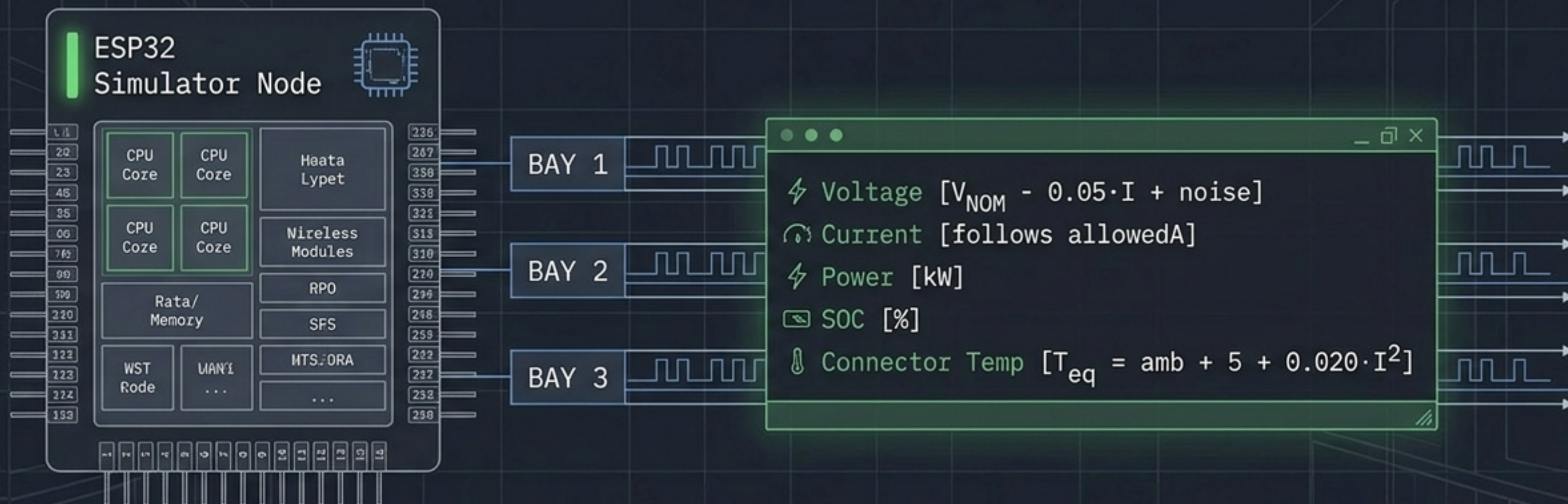
Absolute data privacy

No raw vehicle data leaves the physical site. The cloud is utilized purely for visibility, never for physical control.

EXECUTION STACK TECHNOLOGICAL TRIAD



SENSE PHASE TECHNICAL SCHEMATIC



The Continuous SENSE Loop: Every one second, the firmware simulator calculates electrical voltage sag, CC/CV charging curves, and exponential thermal rise models ($\tau \approx 1.5$ min), generating synthetic but physically accurate raw telemetry.

Model A: Arrival Probability

Will an EV plug in during the next 15 minutes?



Algorithm: Logistic Regression



Scale: Trained on 80,000 synthetic samples

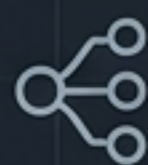


Features: 6 dimensions (time, busy rate, peak hours)

ROC-AUC: 0.79

Model B: Session Duration

How many minutes will this specific session take?



Algorithm: Decision Tree Regressor (Max Depth 8, 511 nodes)



Scale: Trained on 60,000 synthetic samples

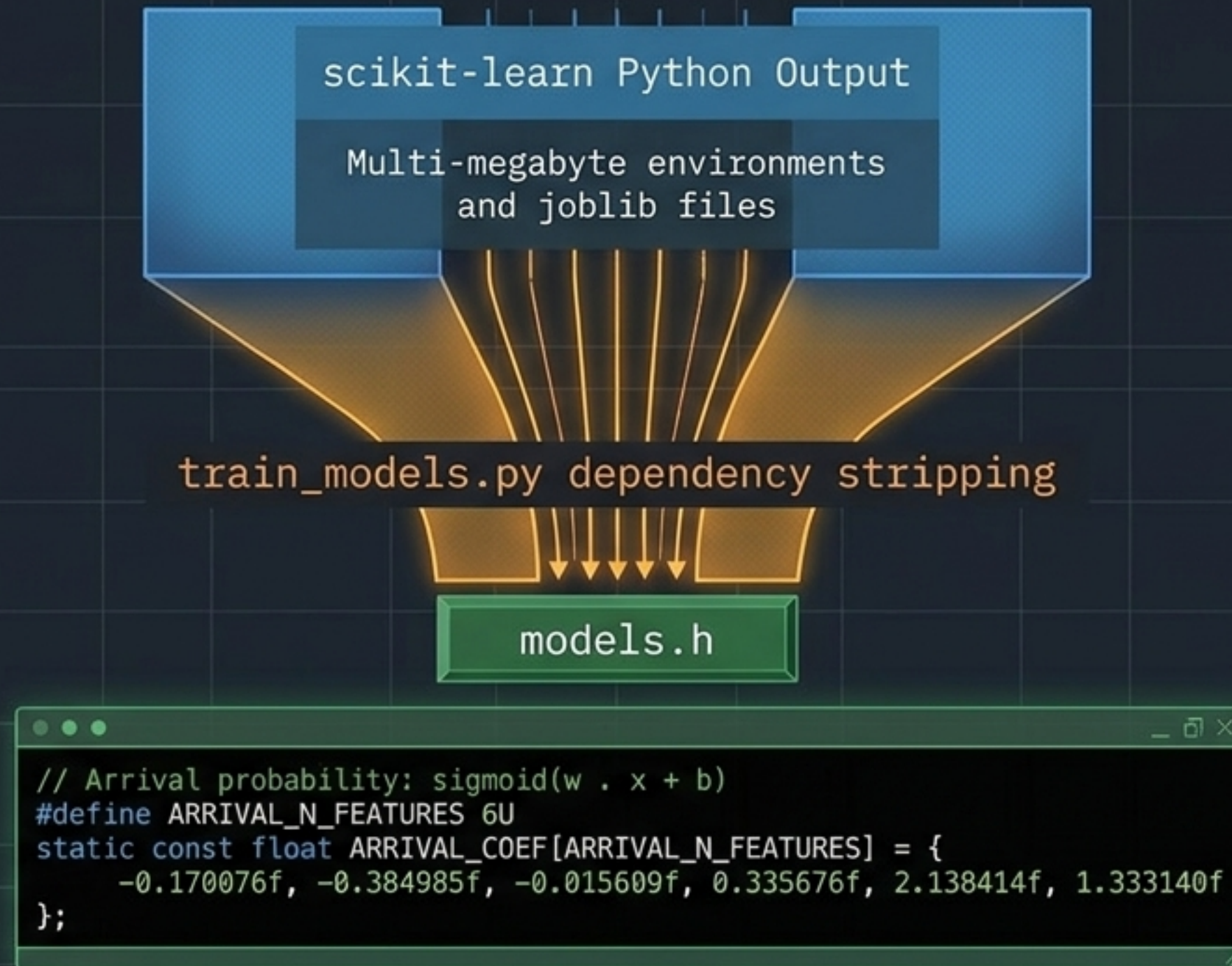


Features: 6 dimensions (capacity, SoC, temperatures)

MAE: ~43 min

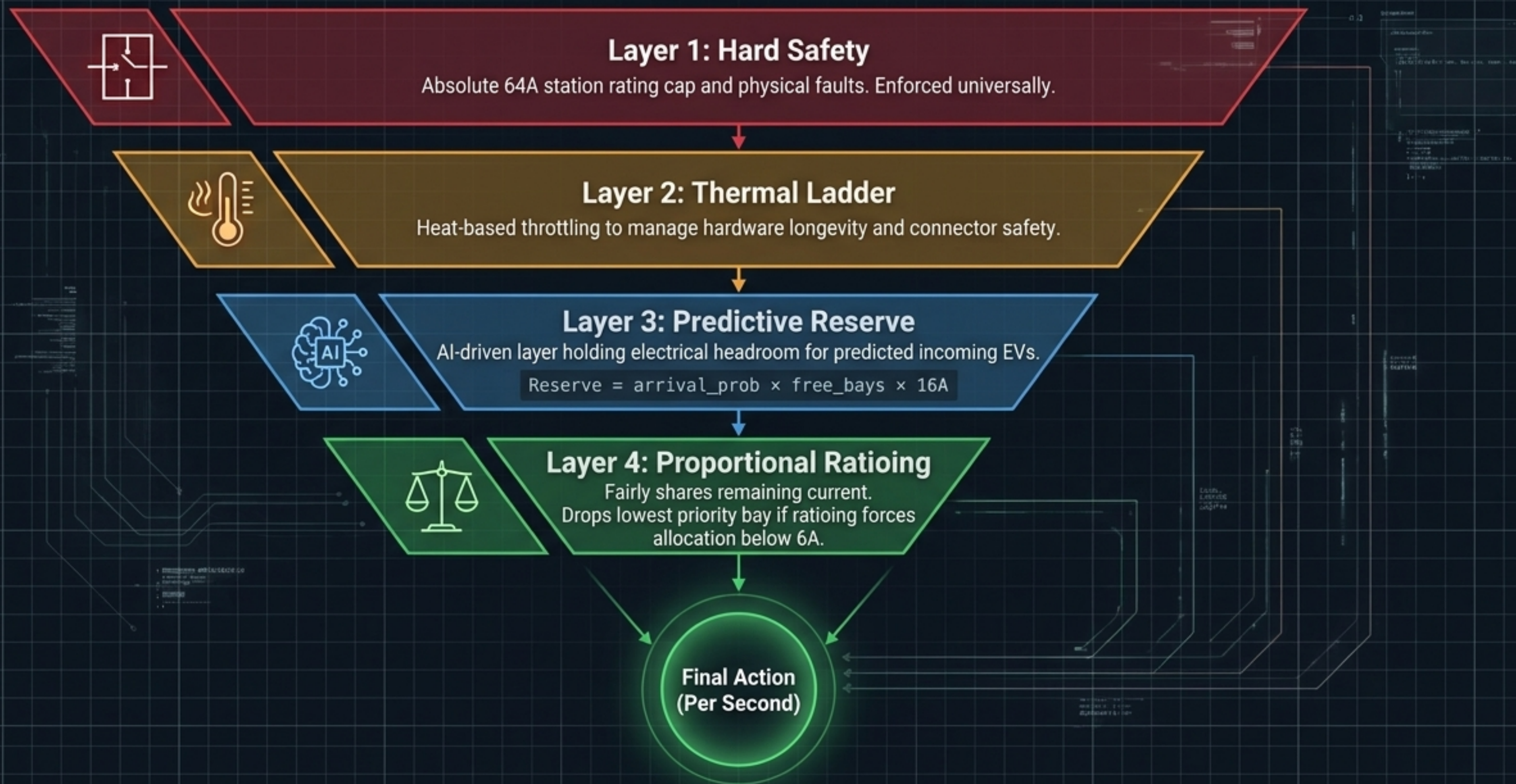
R²: 0.91

DEPLOYMENT PIPELINE: FROM HEAVY ML TO EDGE LOGIC

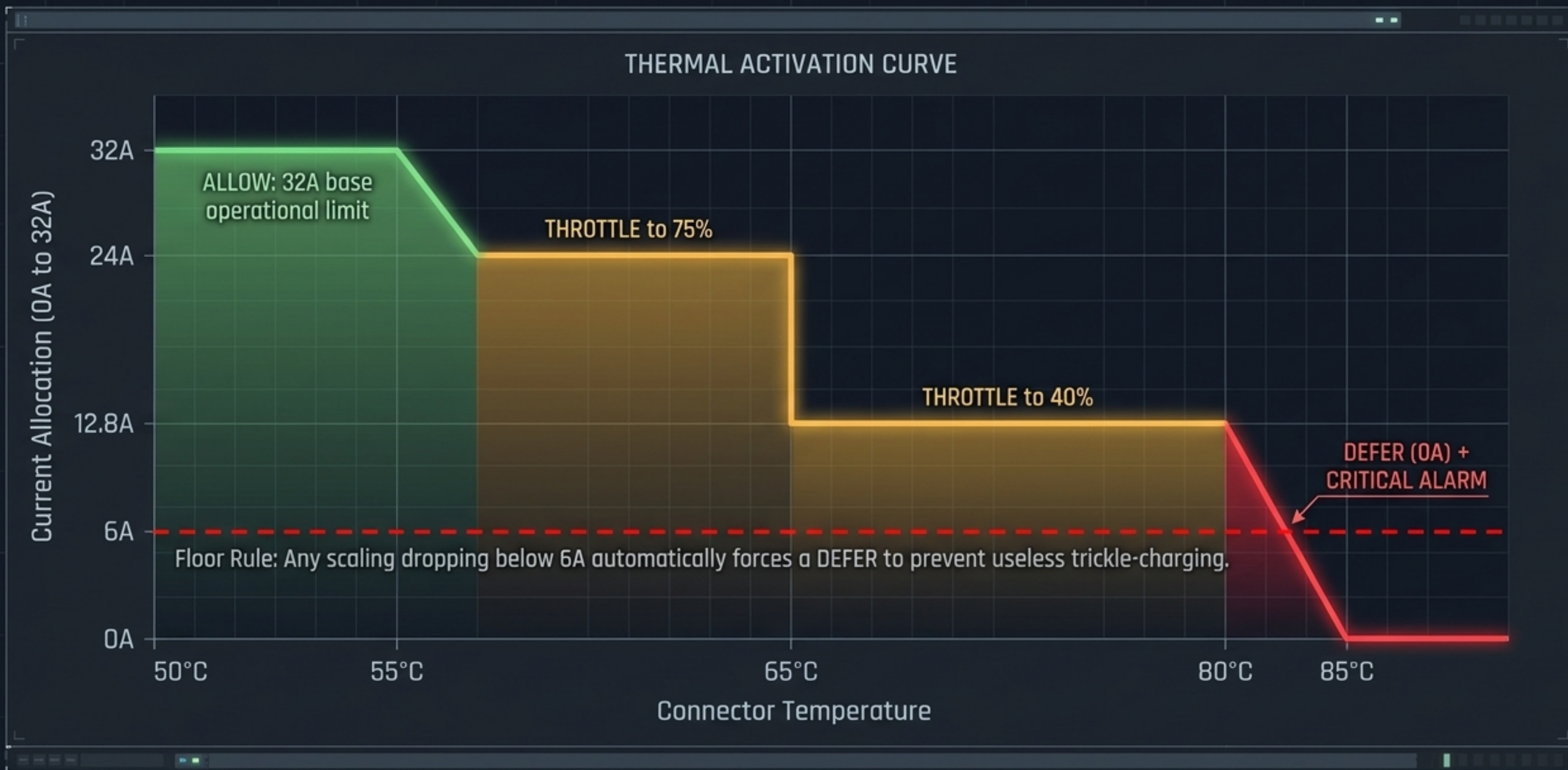


Zero ML libraries on the device. Microsecond execution on a \$10 ESP32.
No TensorFlow Lite. No cloud inference latency.

OPTIMIZER PRIORITY: CASCADING FILTER HIERARCHY



THERMAL-CURRENT STEP-DOWN LADDER: PRECISE THROTTLING FOR CONNECTOR SAFETY



OPTIMIZATION STRATEGY: DYNAMIC TRAFFIC-LIGHT GAUGE SYSTEM



ALLOW

Up to 32A allocation. Safe thermal and electrical headroom.



THROTTLE

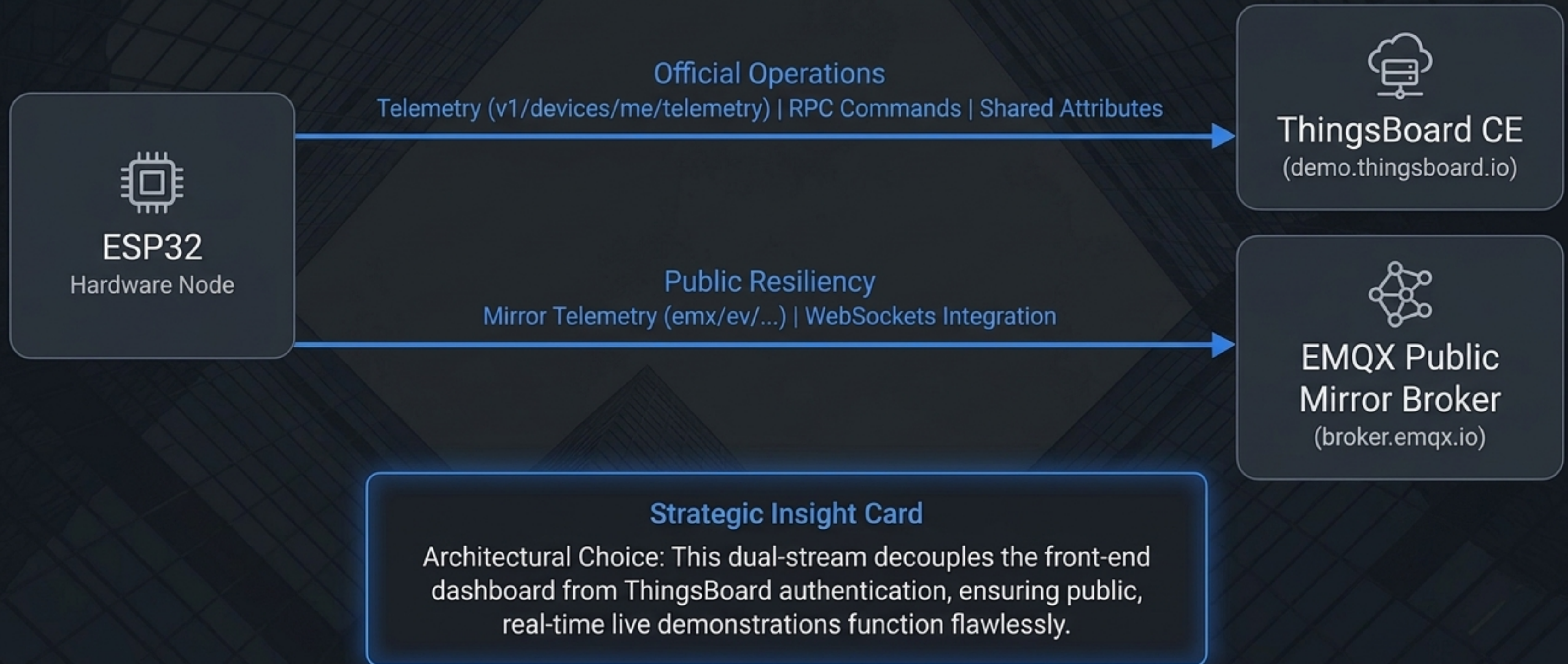
Reduced current allocation. Dynamic throttling to manage connector heat build-up or share limited feeder capacity.



DEFER

0A paused session. Strictly enforced to prevent crossing the 64A feeder limit or 80°C critical thermal threshold.

DUAL-STREAM CLOUD ARCHITECTURE: RESILIENCY THROUGH DECOUPLING



SMART CHARGING STATION: LIVE DASHBOARD INTERFACE

1
Station Load SVG
& AI Arrival Charts



2
Bay Cards:
SoC, Telemetry
& Decision Chips

3
Manual Control Overrides



Offline Fallback Capability
(js/demo.js)

If the MQTT connection drops for 12 seconds, the dashboard flawlessly switches to 'DEMO FEED'.

A full JavaScript port of the entire station, optimizer, and AI takes over locally. The presentation never fails.

SYSTEM VALIDATION & STANDARDIZATION: PROVEN INTEGRITY & UNIVERSAL DATA CONTRACT

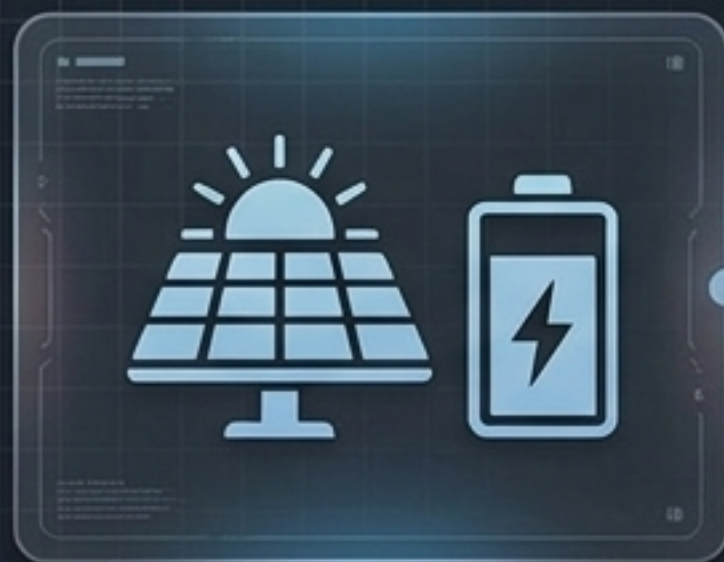
```
[PASS] 14/14 checks passed
[PASS] Tree export matches sklearn (max err 0.0000 min)
[PASS] Hard limit respected (worst 64.0A vs limit 64A)
[PASS] Critical temp defers bay
  Learning export matches and Python simulator
Running regression suite...
  Learning export matches and Python simulator
Execution time: 4.5s
```

```
{
  'station_load_pct': <number>
  'arrival_probability_pct': <number>
  'bay1_decision': <'ALLOW' | 'THROTTLE' | 'DEFER'>
  'alarm_codes': 'TEMP_HIGH | OVERLOAD'
}
```

Flat-Key JSON Schema for
Cross-Platform Interoperability

Universal Data Contract: One identical schema governs the C++ firmware, the Python simulator, the offline JavaScript engine, and the front-end dashboard.

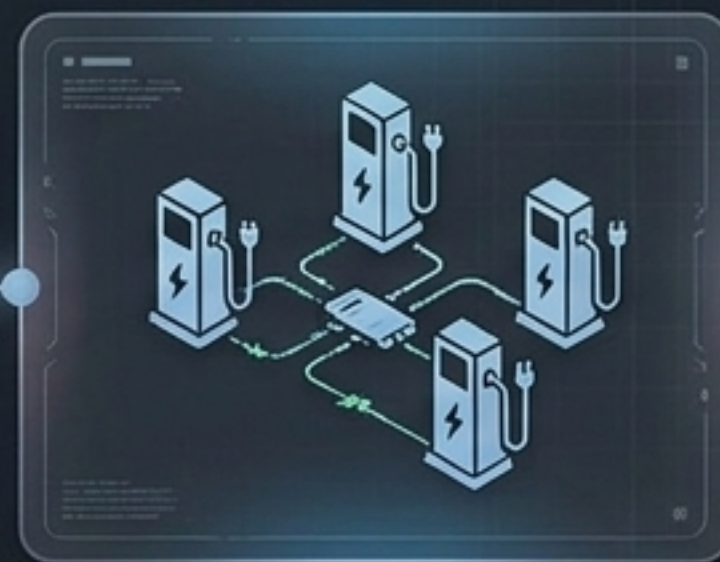
SCALING THE EDGE: BEYOND A SINGLE FEEDER



Solar & BESS Inputs:
Renewable-aware **ALLOW/THROTTLE**
limits based on real-time grid
generation.



Time-of-Use (ToU) **Economics**:
Cheapest-slot scheduling and
grid-peak cost avoidance.



Fleet Orchestration: Multi-Station
MQTT Coordination for managing
fleets of feeders and **integrating**
with commercial OCPP hardware.

Smart EV charging is not only about supplying power—it is about predicting demand, protecting infrastructure, and allocating constrained energy intelligently at the edge.